

Dipak B Patil

Aspiring AIML Engineer

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SUMMARY

Detail-oriented Machine Learning Engineer with a strong foundation in Python, SQL, and core Machine Learning concepts, developed through academic coursework and self-driven learning. Skilled in exploratory data analysis (EDA) and writing analytical queries to extract insights from datasets. Working knowledge of libraries such as NumPy, Pandas, and Scikit-learn, with growing exposure to TensorFlow and PyTorch. Strong analytical and problem-solving skills, with a structured, logical approach to tackling technical challenges. Eager to apply my knowledge in a real-world AI/ML environment and grow as a professional.

EDUCATION

B.Tech – Computer Science & Engineering (Artificial Intelligence & Machine Learning)

2022 – 2026

G.H. Raison College of Engineering and Management, Jalgaon

TECHNICAL SKILLS

- Programming Languages : Python
- Python Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, PyTorch, TensorFlow, LangChain
- Data Analytics and Statistics: Data Cleaning, Preprocessing, EDA, Feature Engineering, Statistical Analysis, Advanced Statistics
- Database & Query: SQL (MySQL), Joins, Subqueries, Aggregations, Stored Procedure
- Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, Feature Engineering
- Tools & Technologies: Streamlit, Git/GitHub, Jupyter Notebook, Google Colab
- Reporting & Analysis: Dashboarding, Ad-hoc Reporting, Data Storytelling

PROJECTS

Text Emotion & Toxicity Detection | Machine Learning, NLP 

Model Building, Text Preprocessing

- Developed an NLP-based system to classify text into emotion categories (joy, anger, sadness, fear) and detect toxic content (hate speech, insults, threats) using ML and DL models.
- Preprocessed text data (tokenization, lemmatization, TF-IDF embeddings) and trained multiple models (LSTM, SVM,) to optimize accuracy and F1-score.
- Deployed the model as a Streamlit, enabling real-time emotion and toxicity analysis for content moderation use cases.

AI-Powered Road Damage Detection using Drone Imagery | ML & DL 

Python, PyTorch, OpenCV, Computer Vision, Object Detection

- Developed an AI-powered road damage detection system using PyTorch and Computer Vision to identify cracks, potholes, and surface defects from high-resolution drone-captured aerial images with accurate object localization.
- Built and trained a deep learning-based object detection model, incorporating image preprocessing, data augmentation, and transfer learning techniques to improve detection accuracy across different road conditions.
- Evaluated the model using Precision, Recall, F1-Score, IoU, and Mean Average Precision (mAP), and optimized its performance through hyperparameter tuning and validation on annotated datasets.

AI Chatbot using LangChain, Hugging Face & Streamlit | LLM 

Generative AI Chatbot with LLM Integration and Structured Output Parsing

- Built a LangChain-based AI chatbot using Hugging Face/NVIDIA LLMs and Streamlit for an interactive user interface.
- Implemented prompt templates and output parsers to generate structured and reliable model responses.
- Integrated LLM APIs and embeddings to demonstrate practical Generative AI and NLP application development.

Hotel Booking Cancellation Analysis | Python, EDA, Power BI 

Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit | DAX Query, Conditional Columns

- BI Dashboard link 
- Performed comprehensive Exploratory Data Analysis (EDA) on 119K+ hotel booking records to identify booking patterns and cancellation trends.
- Cleaned and pre-processed data by handling missing values, duplicates, and data inconsistencies to ensure analysis accuracy.
- Conducted univariate, bivariate, and multivariate analyses to uncover relationships between lead time, hotel type, customer segments, ADR, and cancellations.

Smart-Wallet: Multi-User Personal Finance Web App 

Python, Object-Oriented Programming (OOP), HTML, CSS

- Developed Smart-Wallet, a secure multi-user personal finance web application with authenticated user-specific digital wallets and complete data isolation.
- Implemented features to add balance, record expenses, and view transaction history through a bank-style passbook interface.
- Enabled monthly financial report generation to support expense tracking, budgeting, and better financial decision-making.

CERTIFICATES

- Python Programming 
- Exploratory Data Analysis 
- Power BI 
- MySQL 